

Pre-Requisites

- Basic understanding of machine learning concepts such as features, labels, training, and evaluation.
- Familiarity with classical ML algorithms like Logistic Regression or Decision Trees.
- Awareness of mathematical concepts such as linear combinations, gradients, and activation functions (intuitive understanding is sufficient).
- No coding experience required, but prior exposure to model building in Python or similar tools will be helpful.

Session Summary

This 3-hour live session introduces learners to the fundamental concepts of **Artificial Neural Networks (ANNs)** — the building blocks of modern deep learning. Learners will explore how neural networks are inspired by the human brain, understand their architecture (neurons, layers, and activations), and learn how information flows through the network. The instructor will explain key ideas such as forward and backward propagation in an intuitive, non-mathematical manner.

Note: This is a concept-driven session. Learners are encouraged to ask questions, participate in discussions, and relate neural network concepts to traditional ML models they have studied earlier.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Explain the motivation behind neural networks and how they differ from traditional ML models.
- Describe the basic architecture of a neural network — including input, hidden, and output layers.
- Understand how neurons process information through weights, biases, and activation functions.
- Interpret the concepts of forward propagation and backward propagation at a high level.
- Recognize the role of hidden layers in learning complex, non-linear relationships.
- Relate neural network concepts to real-world applications across domains such as image, speech, and text processing.

Brief Agenda

1. **Introduction & Motivation** (10 mins)
Overview of session goals and why neural networks are needed. Discuss how classical ML struggles with complex, non-linear data patterns.
2. **Understanding the Neuron** (15 mins)
Explain the structure and working of an artificial neuron — inputs, weights, bias, and activation. Relate it to the biological neuron.
3. **Network Architecture & Layers** (20 mins)
Describe input, hidden, and output layers. Discuss how stacking layers enables learning of complex representations.
4. **Forward & Backward Propagation** (25 mins)
Explain, at a conceptual level, how data flows through the network (forward pass) and how the model learns by adjusting weights (backward pass).

5. **Activation Functions** (20 mins)
Introduce ReLU, Sigmoid, and Tanh. Explain why non-linearity is essential for learning complex mappings.
6. **Break** (10 mins)
7. **Practical Intuition: From ML to NN** (20 mins)
Compare how Logistic Regression extends into a simple neural network. Discuss intuition behind learning multiple layers.
8. **Learner Discussion & Q&A** (15 mins)
Encourage learners to share their understanding and real-world scenarios where neural networks could be applied.
9. **Conclusion & What's Ahead** (5 mins)
Recap key learnings and preview the next session on CNNs and RNNs.

Detailed Agenda

Sample Start (T+)	Duration (min)	Slot Title	Objective
00:00	15	Introduction & Motivation	Introduce the goals of the session and the need for neural networks. Explain how traditional ML methods fall short for complex, non-linear problems and how neural networks overcome these limitations.
00:15	20	Understanding the Neuron	Explain the structure and functionality of an artificial neuron, including inputs, weights, bias, and activation. Relate the artificial neuron to its biological inspiration.
00:35	25	Network Architecture & Layers	Discuss how neurons combine to form networks. Explain the role of input, hidden, and output layers, and demonstrate how depth enables learning of complex abstractions. Use visual diagrams or an interactive demo for better understanding.
01:00	25	Forward & Backward Propagation	Walk through how data flows through the network (forward pass) and how errors are minimized (backpropagation). Explain gradient descent intuitively without heavy mathematics.
01:25	10	Break	Short break for learners to refresh and process initial concepts.
01:35	25	Activation Functions	Discuss the role of activation functions in introducing non-linearity. Cover common functions (Sigmoid, Tanh, ReLU) and demonstrate their impact visually on data representation and learning.
02:00	20	From Logistic Regression to Neural Networks	Show how Logistic Regression can be seen as a one-layer neural network. Explain how adding layers allows models to learn more complex decision boundaries.
02:20	20	Practical Intuition & Real-World Relevance	Connect concepts to real-world applications—such as image classification, speech recognition, and recommendation systems. Emphasize why neural networks are foundational to deep learning.
02:40	15	Learner Discussion & Q&A	Encourage learners to share their takeaways and identify potential use cases of neural networks in their professional or academic contexts. Address conceptual doubts and clarify key ideas.
02:55	5	Conclusion & What's Ahead	Summarize the key points of the session. Reinforce the relevance of neural networks as a stepping stone to CNNs and RNNs, and preview upcoming sessions on these architectures.

Conclusions

By the end of this live session, learners will have developed a strong conceptual understanding of how neural networks function as the foundation of deep learning. They will be able to explain the role of neurons, layers, weights, and activation functions in building predictive models and will understand how forward and backward propagation enable learning from data. Learners will also be able to relate these concepts to classical ML models like Logistic Regression and appreciate how neural networks extend these ideas to capture complex, non-linear relationships. This foundational knowledge will prepare them for upcoming sessions on specialized architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs).